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ORACLE GOLDENGATE POCKET GUIDE

DATABASES | CLOUD | MSP| ENGINEERED SYSTEMS | REPLICATION | CONSULTING | INFRASTRUCTURE | SECURITY

Oracle GoldenGate Processes

Manager
It is required for Stop and Start GoldenGate Extract, Pump and Replicat processes

Extract
It reads database from online redo log files and optionally from archive log files and write data to local or remote trail files

Data Pump Extract
It reads data from local trail file and writes data to remote trail over the TCP/IP network. It is an optional process but highly recommended

Collector
It receives data from Pump Extract and writes data to remote trail files

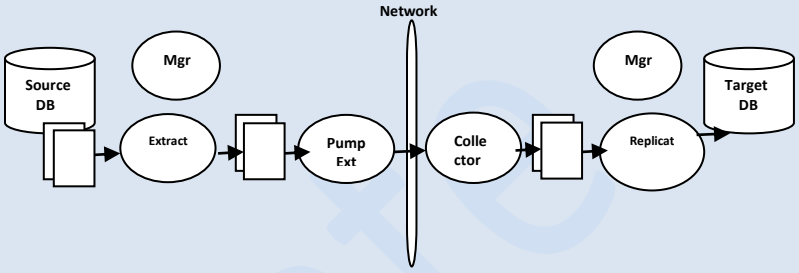
Replicat
It reads data from remote trail files and write to the database one transaction at a time

Trails and Files
Trails files are required for continuous data replication. These files are canonical structured files which only GoldenGate process can read

Checkpoints
It provides the transaction recovery in case GoldenGate processes stopped abnormally.

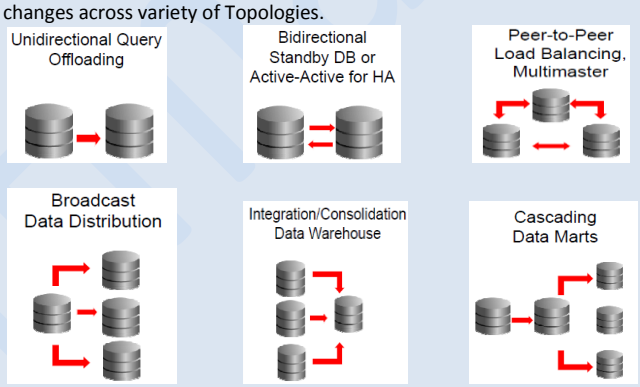
Wallet
It is a secure common storage for User IDs and Passwords. It is highly recommended to use wallet for secure credentials

Oracle GoldenGate Architecture



This diagram explains the logical architecture of an Oracle GoldenGate online replication.

GoldenGate Topologies
Oracle GoldenGate modular architecture provides the ability to extract DML & DDL changes across variety of Topologies.



Oracle GoldenGate Utilities

GGSCI
GoldenGate Software Command Line Interface (GGSCI) is used to add, stop, start and manage GoldenGate Processes.

DEFGEN
This utility produces file containing a definition of the layouts of the source tables

LOGDUMP
It enables you to search or display information stored in the GoldenGate trails and file

REVERSE
It reorders operations within GoldenGate trail files in reverse sequence

KEYGEN
It is used to generate random hex keys. Needed only if you do not use wallet

INITIAL LOAD
- **File to Replicat**

Extract process write data to trail files and Replicat loads data using SQL on target DB
- **File to Database Utility**

Extract process write data to text files and Replicat loads data using DB utility on target
- **Direct Load**

Extract process gives data directly to Replicat & Replicat loads data using SQL on target
- **Direct Bulk Load**

Extract process gives data directly to Replicat & Replicat loads data using DB utility

Oracle GoldenGate 11g/12c Installation

Download latest Oracle GoldenGate 11g/12c from https://support.oracle.com

- **Oracle GoldenGate 11g Installation**

Login as OGG software owner
$ cd /ggs/home
$ $ p18918679_1121017_Linux-x86-64.zip
$ tar -xvof *.tar
$ ./ggsci
GGSCI> create subdirs
GGSCI> info all

- **Oracle GoldenGate 12c Installation**

Login as OGG software owner
$ unzip p27111516_122022_Linux-x86-64.zip
$ cd fbo_ggs_Linux_x64_shiphome/Disk1/response/
$ cp oggcore.rsp /u01/stage/
$ cd /u01/stage/

Open the response and edit the following parameters
$ vi oggcore.rsp
INSTALL_OPTION=ORA11g
SOFTWARE_LOCATION=/u01/app/oracle/product/ogg/12.2.0.22/gghome
START_MANAGER=true
MANAGER_PORT=7809
DATABASE_LOCATION=/u01/app/oracle/product/12.2.0/dbhome
INVENTORY_LOCATION=/u01/app/oraInventory
UNIX_GROUP_NAME=oinstall

$ ./runInstaller -silent -responseFile /u01/stage/oggcore.rsp

Oracle GoldenGate One-Way Replication Setup – Classic

High-levels to configure Oracle GoldenGate one-way replication

- **Install OGG software on source and Target**

- **Configure OGG Manager process on source and target**

```
GGSCI> Edit params mgr
PORT 7809
PURGEOLDEXTRACTS /ggs/home/dirdat/*, USECHECKPOINTS, MINKEEPDAYS 2
```

- **Enable DB Supplemental logging on**

```
SQL> alter database add supplemental log data;
SQL> select supplemental_log_data_min from v$database;
```

- **Add Trandata on tables**

```
GGSCI> dblogin userid ggs, password xxxxx
GGSCI> Add trandata SCOTT.*
```

- **Configure Extract & Pump Extract Processes on source**

```
GGSCI> edit params ext
EXTRACT intext
USERID ggs, PASSWORD xxxxx
EXTTRAIL ./dirdat/et
TABLE SCOTT.*;

GGSCI> add extract test, tranlog, begin now
GGSCI> add exttrail /ggs/home/dirdat/ex2c157a/et, extract ext, megabytes 100
GGSCI> start ext

GGSCI> edit params pext
EXTRACT pext
RMTHOST 192.168.10.1, MGRPORT 7809
RMTTRAIL /ggs/home/dirdat/rt
PASSTHRU
TABLE SCOTT.*;

GGSCI> add extract pext, exttrailsource /ggs/home/dirdat/et
GGSCI> add rmttrail /ggs/home/dirdat/rt, extract ptest, megabytes 100
GGSCI> start pext
```

- **Perform Initial using your desired method (Expdp/impdp, RMAN, and so on)**

```
$ expdp userid = ggs/xxxx flashback_scn = xxxxx directory = EXP_DIR dumpfile =
expdp_src.dmp logfile = expdp_src.log status=60 schemas = SCOTT

$ impdp userid = ggs/xxxx directory = IMP_DIR dumpfile = expdp_src.dmp logfile =
impdp_tgt.log status=60 remap_schemas = SCOTT:SCOTT
```

- **Create checkpoint table**

```
$ ./ggsci

GGSCI> dblogin userid ggs, password xxxx
GGSCI> add checkpointtable ggs.chkpoint_table
GGSCI> edit params ./GLOBALS

CHECKPOINTTABLE GGS.CHKPOINT_TABLE

GGSCI> exit
```

- **Configure Replicat on Target**

```
GGSCI> edit params rep
REPLICAT REP
USERID ggs, PASSWORD xxxxxx
ASSUMTARGETDEFS
MAP SCOTT.*, TARGET SCOTT.*;

GGSCI> dblogin userid ggs, password xxxxx
GGSCI> add replicat rep, exttrail /ggs/home/dirdat/rt
GGSCI> start replicat
GGSCI> info rep
```

Oracle GoldenGate One-Way Replication Setup – Integrated (12c)

11.2.0.4 and above

```
SQL> alter system set ENABLE_GOLDENGATE_REPLICATION=true;
```

Create Datastore

```
GGSCI> create wallet
GGSCI> add credentialstore
GGSCI> alter credentialstore add user ggs
GGSCI> info credentialStore
GGSCI> dblogin useridalias ggs

GGSCI> add trandata SCOTT.*  ALLCOLS
```

Integrated Extract

```
SQL> EXEC DBMS_GOLDENGATE_AUTH.GRANT_ADMIN_PRIVILEGE(grantee =>
'GGUSER');

$ cd $GGS_HOME
$ ./ggsci
GGSCI> EDIT PARAMS intext
EXTRACT intext
dblogin useridalias ggs
RMTHOST 192.168.10.1, MGRPORT 7809
TRANLOGOPTIONS INTEGRATEDPARAMS (MAX_SGA_SIZE 100)
RMTTRAIL ./dirdat/ie
TABLE SCOTT.*;
GGSCI> ADD EXTRACT intext, INTEGRATED TRANLOG, BEGIN NOW
GGSCI> dblogin useridalias ggs
GGSCI> REGISTER EXTRACT intext, DATABASE
GGSCI> ADD RMTTRAIL ./dirdat/ie, EXTRACT intext
GGSCI> start extract intext
```

Login to the source database and query the $GOLDENGATE_CAPTURE view to get the information about the integrated extract.

```
SQL> select sid, serial#,CAPTURE_NAME,STATE, SGA_USED from
V$GOLDENGATE_CAPTURE;
```

Integrated Replicat

```
$ cd $GGS_HOME
$ ./ggsci

GGSCI> dblogin useridalias ggs
GGSCI> add checkpointtable ggs.chkpoint_table
GGSCI> edit params ./GLOBALS
CHECKPOINTTABLE GGS.CHKPOINT_TABLE

GGSCI> exit

GGSCI> Edit Param rint
Replicat rint
DBOPTIONS INTEGRATEDPARAMS(parallelism 6)
assumetargetdefs
discardfile ./dirrpt/rint.dsc, Purge
dblogin useridalias ggs
Map scott.*, target scott.*;

GGSCI> dblogin useridalias ggs
GGSCI> add Replicat rint Integrated  exttrail ./dirdat/ie
GGSCI> Start Replicat rint
GGSCI> info all

SQL>  connect / as sysdba
SQL>  column replicat_name format a30
SQL>  column server_name format a30
SQL> select replicat_name,server_name from DBA_GOLDENGATE_INBOUND;
```

Oracle GoldenGate Patching (12c)

It is assumed that base OGG software version 12.2.0.1.0 or 12.2.0.1.1 or 12.1.2.1.0 or 12.1.2.1.1 is already installed.

Steps:
Download and unzip patch in staging area
$ cd /stage
$ unzip *.zip → unzip patch

$ cd $GG_HOME
$ ./ggsci
GGSCI> version

$ export ORACLE_HOME=/u01/app/oracle/product/ogg/12.2.0.1.1/ogghome1
$ export PATH=$PATH:/u01/app/oracle/product/12.2.0/orcl/OPatch
$ opatch version
$ opatch lsinventory

$ ./ggsci
GGSCI> STOP ER *
GGSCI> STOP MANAGER
$ cd /stage/<patch_number>

$ opatch apply

$ export ORACLE_HOME=/u01/app/oracle/product/11.2.0/orcl

$ ./ggsci
GGSCI> START MANAGER
GGSCI> START ER *

Oracle GoldenGate Upgrade (12c)

It is assumed that we are upgrade OGG from 12.2.0.1.0 to 12.2.0.1.1

Steps:
Download and unzip patch in staging area
$ cd /stage
$ unzip *.zip → unzip patch

$ cd $GG_HOME
$ ./ggsci
GGSCI> stop er *
GGSCI> stop mgr!

$ cd /u01/app/oracle/product/ogg/12.2/ogghome1
$ cd ..
$ cp -pR ogghome1 ogghome1_bkp

$ cd /stage/xxx/Disk1
$ ./runInstaller → follow the on screen details to complete the installation

$ cd $GG_HOME
$ ./ggsci -v

$ cd /u01/app/oracle/product/ogg/12.2/ogghome1_bkp
cp -pR dir* $GG_HOME

$ ./ggsci
GGSCI> START MANAGER
GGSCI> START ER *

Oracle GoldenGate Initial Load Methods

Direct Load Method

- **Create initial data load extract**

$ cd $GG_HOME
$ ./ggsci
GGSCI> ADD EXTRACT ELOAD, SOURCEISTABLE
GGSCI> INFO EXTRACT ELOAD, TASKS

GGSCI> EDIT PARAMS ELOAD
EXTRACT ELOAD
USERID ggs, PASSWORD ggs
RMTHOST 10.10.10.2, MGRPORT 7809
RMTTASK REPLICAT, GROUP RLOAD
TABLE SCOTT.*;

GGSCI> ADD EXTRACT ELOAD, SOURCEISTABLE

- **Create initial data Replicat process on target**

$ cd $GG_HOME
$ ./ggsci
GGSCI> ADD REPLICAT RLOAD, SPECIALRUN
GGSCI> INFO RLOAD, TASKS

GGSCI> EDIT PARAMS RLOAD
REPLICAT RLOAD
USERID ggs, Password ggs
DISCARDFILE ./dirdat/rustarc.DSC, PURGE
BATCHSQL
MAP SCOTT.*, TARGET SCOTT.*;

GGSCI> ADD REPLICAT RLOAD, SPECIALRUN

- **Start Initial Load on source only**

GGSCI> start eload
GGSCI> info eload
GGSCI> info rload → check on target

File to Replicat Method

- **Create initial data load extract**

$ cd $GG_HOME
$ ./ggsci

GGSCI> Edit Param eload
SOURCEISTABLE
USERID ggs, PASSWORD ggs
RMTHOST 10.10.10.2, MGRPORT 7809
-- Below rmtfile command will create 50 files of size 2GB each on target server
RMTFILE ./dirdat/el, maxfiles 50 MEGABYTES 2048
TABLE SCOTT.*;
GGSCI> exit

$ ./extract paramfile dirprm/eload.prm reportfile dirrpt/eload.rpt
$ ps -ef|grep eload
$ cat dirrpt/eload.rpt | more

- **Create initial data Replicat process on target**

$ cd $GG_HOME
$ ./ggsci
GGSCI> add rep rload, exttrail ./dirdat/el, nodbcheckpoint

GGSCI> EDIT PARAMS rload
Replicat RLOAD
USERID ggs, Password ggs
DISCARDFILE ./dirdat/rustarc.DSC, PURGE
DBOPTIONS SUPPRESSTRIGGERS
DBOPTIONS USEREPLICATIONUSER
BATCHSQL
MAP SCOTT.*, TARGET SCOTT.*;

GGSCI> start replicat rload

$ ps -ef|grep rload
$ cat dirrpt/rload.rpt | more

Useful Oracle GoldenGate Parameter files

Manager Parameter File

```
PORT 7809
DYNAMICPORTLIST 7810-7830
USERID ggs, PASSWORD abcbd!@#$1234, & ENCRYPTKEY DEFAULT
PURGEOLDEXTRACTS /ggs/home/dirdat/* , USECHECKPOINTS, MINKEEPHOURS 48
AUTOSTART ER *
AUTORESTART ER *, RETRIES 3, WAITMINUTES 10, RESETMINUTES 720
DOWNREPORTMINUTES 15
DOWNCRITICAL
LAGCRITICALSECONDS 10
LAGINFOMINUTES 0
LAGREPORTMINUTES 15
```

Extract Parameter File

```
EXTRACT EXT_TEST
SETENV (ORACLE_HOME = "/u01/app/oracle/product/12.0.1/db")
SETENV (NLS_LANG="AMERICAN_AMERICA.AL32UTF8")
USERID ggs, PASSWORD "abcd!@#$1234", & ENCRYPTKEY DEFAULT
EXTTRAIL /ggs/home/dirdat/et
DISCARDFILE /ggs/home/dirrpt/ext_test.dsc, megabytes 100 append
DISCARDROLLOVER AT 02:00 ON SUNDAY
REPORTCOUNT EVERY 10000 RECORDS, RATE
STATOPTIONS REPORTFETCH
STATOPTIONS RESETREPORTSTATS
REPORT ON SUNDAY AT 01:55
REPORTROLLOVER AT 01:00 ON SUNDAY
TRANLOGOPTIONS EXCLUDEUSER ggs
GETUPDATEBEFORES
DDL INCLUDE MAPPED OBJNAME SCOTT.*
DBOPTIONS ALLOWUNUSEDCOLUMN
EOFDELAYCSECS 10
TABLE SCOTT.*;
TABLEEXCLUDE SCOTT.TEST
SEQUENCE SCOTT.*;
```

Data Pump Extract Parameter File

```
EXTRACT PUMP_TEST
PASSTHRU
RMTHOST 192.168.10.1, MGRPORT 7809, COMPRESS, TCPBUFSIZE 900000
TCPFLUSHBYTES 900000
EOFDELAYCSECS 10
FLUSHCSECS  10
RMTTRAIL /ggs/home/dirdat/rt
TABLE SCOTT.*;
```

Replicat Parameter File

```
REPLICAT REP_TEST
SETENV (ORACLE_HOME = "/u01/app/oracle/product/12.0.1/db")
SETENV (NLS_LANG="AMERICAN_AMERICA.AL32UTF8")
USERID ggs, PASSWORD "abcd!@#$1234", & ENCRYPTKEY DEFAULT
ASSUMETARGETDEFS
DISCARDFILE /ggs/home/dirrpt/rep_test.dsc, megabytes 100 append
DISCARDROLLOVER AT 01:00 ON SUNDAY
DBOPTIONS DEFERREFCONST
BATCHSQL
GROUPTRANSOPS 2000
EOFDELAYCSECS 10
APPLYNOOPUPDATES
REPORTCOUNT EVERY 100000 RECORDS, RATE
REPORT ON SUNDAY AT 00:55
REPORTROLLOVER AT 01:00 ON SUNDAY
STATOPTIONS RESETREPORTSTATS
REPERROR (DEFAULT, EXCEPTION)
REPERROR (DEFAULT2, ABEND)
REPERROR (-1, DISCARD)
REPERROR (1403, IGNORE)
DDL INCLUDE MAPPED
MAP SCOTT.*, TARGET SCOTT.*;
MAPEXCLUDE SCOTT.TEST
```

Oracle GoldenGate Monitoring, Troubleshooting and Tuning

Monitoring and Troubleshooting

```
GGSCI> info all
GGSCI> info mgr
GGSCI> info extract ext
GGSCI> info extract pump
GGSCI> info replicat rep

GGSCI> info extract ext, detail
GGSCI> info extract pump, detail
GGSCI> info replicat rep, detail

GGSCI> send extract ext, status
GGSCI> send extract pump, status
GGSCI> send replicat rep, status

GGSCI> stats extract ext
GGSCI> stats replicat rep

GGSCI> status extract *
GGSCI> status replicat *

GGSCI> info extract int, showch
GGSCI> info replicat rint, showch

GGSCI> lag extract ext
GGSCI> lag replicat rep

GGSCI> view report eint
GGSCI> view report rep

GGSCI> view GGSEVT
```

```
$ view ./dirrpt/einit.dsc
```

showsyntax
Shows Replicat SQL Statement
checkparams
Use to verify syntax

logdump: It enables you to search or display information stored in the GoldenGate trails and file

```
$ logdump
Logdump> ghdr on
Logdump> detail on
Logdump> detail data
Logdump> reclen 200
Logdump> POS 0
Logdump> n
```

Tuning

dynamicresolution
Use this parameter to add a table's attributes to the object record the first time its object ID enter the transaction log
wildcardresolve
Use this parameter if you are using wildcard names
flushsecs
Control the point at which Extract Flushes data from buffer to target
tcpflushbytes
Control the point at which Extract Flushes data from buffer to target
tcpbufsize
Controls the size of the TCP socket buffer
eofdelay
Control how often Extract reads the transaction log or trail file
eofdelaycsecs
Control how often Extract reads the transaction log or trail file
checkpointsecs
Adjust checkpoint for Exadata & Replicat
grouptransops
Controls how many operations are grouped into one transaction
autostart
Control Auto start of the process when manager process starts
bootdelayminutes
Delays Manager start of processing activities on Windows system
passthru
Use this in Pump process if no conversion or filtering is used
batchsql
batches similar SQL statement into arrays
maxtransops
To split large transaction into smaller ones on the target
maxsqlstatements : Replicat maintains cursors for caching SQL statement